

ASSIGNMENT 2 - TASK 1

Report on Optimization Algorithms on Rosenbrock and Ackley Functions

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1. Introduction

This report summarizes the numerical optimization experiments. The notebook investigates two objective functions—Rosenbrock and Ackley function (1D simplified) and applies five optimization methods: Gradient Descent, SGD with Momentum, Adam, RMSprop, and Adagrad. Each optimizer is evaluated at learning rates 0.01, 0.05, and 0.10, with iteration count, execution time, final point, final objective value, and convergence status recorded.

2. Objectives

- Implement and optimize the Rosenbrock function.
- Implement and optimize the Ackley function (1D simplified).
- Compare Gradient Descent, SGD with Momentum, Adam, RMSprop, and Adagrad.
- Investigate how the learning rate affects convergence behavior.
- Record final solution, objective value, iterations, execution time, and status.
- Visualize convergence using objective-value-versus-iteration plots.

3. Optimized value of given non-convex functions

We have used Nelder-Mead method for optimizing the given non-convex functions.

Rosenbrock function, is defined as $f(x,y) = (1-x)^2 + 100(y-x^2)^2$, whose global minimum is at (1,1) with $f=0$.

For Ackley function (1D simplified), $f(x) = -20\exp(-0.2\sqrt{x^2}) - \exp(\cos(2\pi x)) + 20 + e$. The global minimum of the function is at $x=0$ with $f(x)=0$.

4. Optimization Methods

A. Gradient Descent: Updates the parameters directly using the negative gradient direction.

For the **Gradient Descent (GD)** implementation, formula used in our code is:

$$x_{t+1} = x_t - \eta \nabla f(x_t)$$

where:

- x_t = current parameter/point
- x_{t+1} = updated parameter/point
- η = learning rate
- $\nabla f(x_t)$ = gradient of the objective function

B. SGD with Momentum: Maintains a velocity term so that previous update information influences the current step.

For the **SGD with Momentum** implementation, formula used in our code is:

$$v_t = \beta v_{t-1} + (1 - \beta) \nabla f(x_t)$$

$$x_{t+1} = x_t - \eta v_t$$

Where:

- v_t = current velocity
- v_{t-1} = previous velocity
- β = momentum coefficient
- $\nabla f(x_t)$ = current gradient
- x_t = current parameter/point
- x_{t+1} = updated parameter/point
- η = learning rate

C. Adam: Maintains first- and second moment estimates of the gradient and applies bias correction before updating the parameters.

Adam combines **Momentum** and **RMSprop-like adaptive scaling**.

It uses two moving averages:

- First moment m_t
- Second moment v_t

Step 1: First moment

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

where:

$$g_t = \nabla f(x_t)$$

Step 2: Second moment

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

Step 3: Bias correction

Adam corrects the initial bias in the moving averages:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

and

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

Step 4: Parameter update

$$x_{t+1} = x_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

Where:

$$\begin{aligned}\beta_1 &= 0.9 \\ \beta_2 &= 0.999 \\ \epsilon &= 10^{-8}\end{aligned}$$

D. RMSprop: Uses an exponentially weighted moving average of squared gradients to adapt the update magnitude.

RMSprop maintains a moving average of the **squared gradients**.

Step 1: Moving average

$$v_t = \beta v_{t-1} + (1 - \beta) g_t^2$$

where:

$$g_t = \nabla f(x_t)$$

Step 2: Parameter update

$$x_{t+1} = x_t - \eta \frac{g_t}{\sqrt{v_t} + \epsilon}$$

E. Adagrad: Accumulates squared gradients over time and divides the gradient by the square root of this accumulated quantity.

Step 1: Accumulate squared gradients

$$G_t = G_{t-1} + g_t^2$$

Step 2: Parameter update

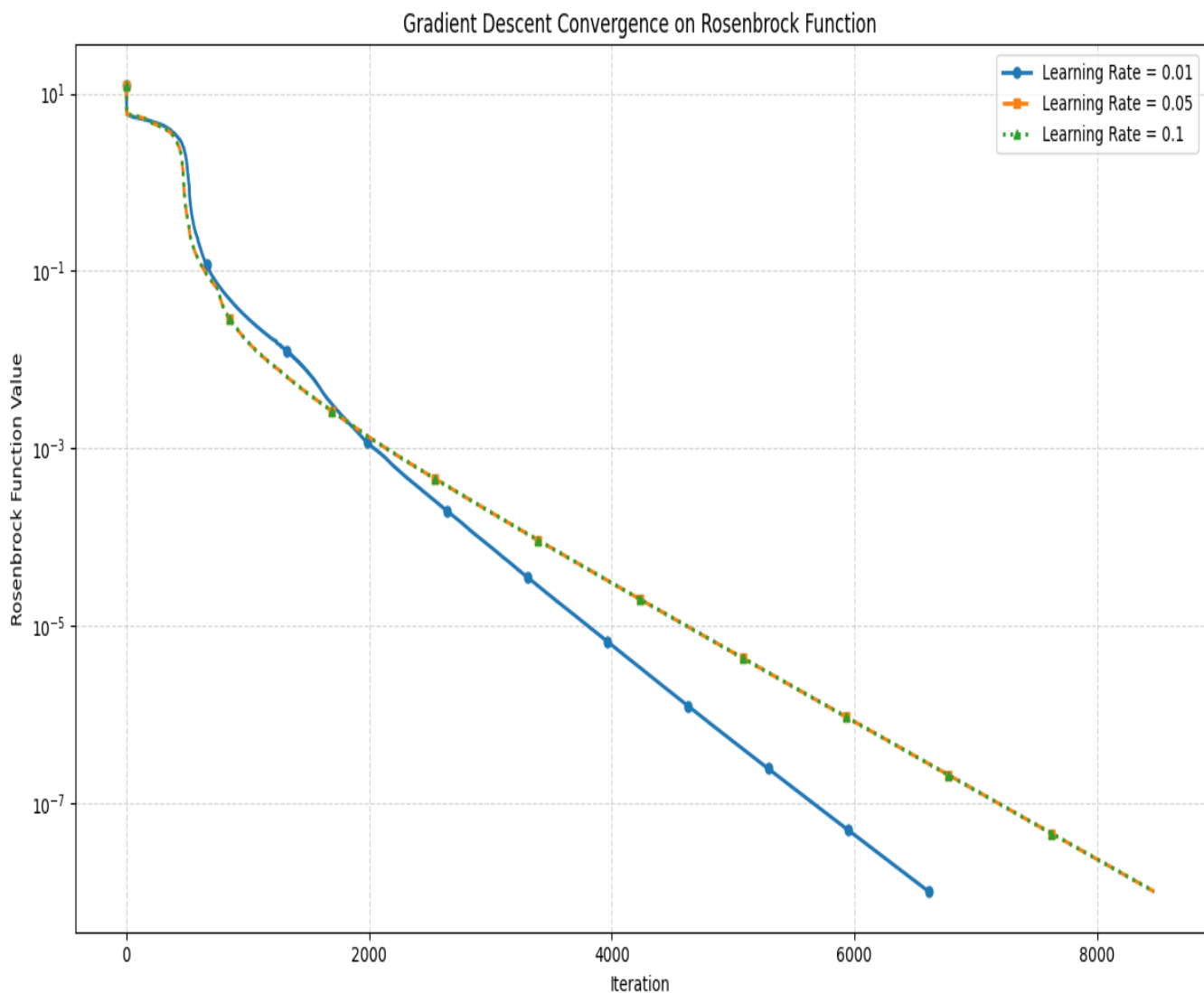
$$x_{t+1} = x_t - \eta \frac{g_t}{\sqrt{G_t} + \epsilon}$$

5. Rosenbrock Results

Firstly we starts the Rosenbrock optimizer experiments from $[-1.5, 2.0]$ and evaluates the three required learning rates $\eta = 0.01, 0.05, 0.1$.

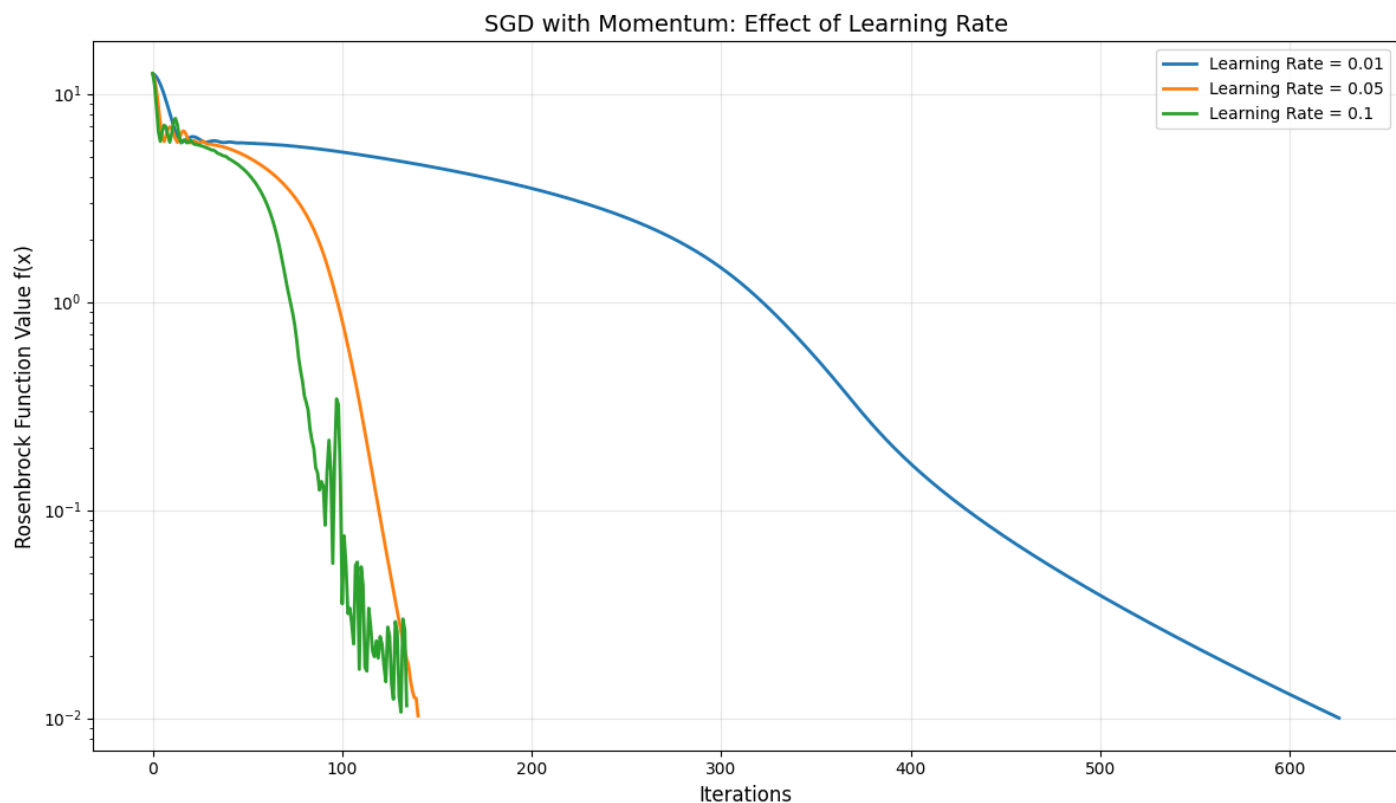
Gradient Descent — Rosenbrock

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[0.9999003489869744, 0.999800157061169]	9.960667e-09	6614	0.465102	Converged
0.05	[0.999900243590124, 0.9997998386440772]	9.994702e-09	8470	0.549710	Converged
0.10	[0.999900243590124, 0.9997998386440772]	9.994702e-09	8470	0.597312	Converged



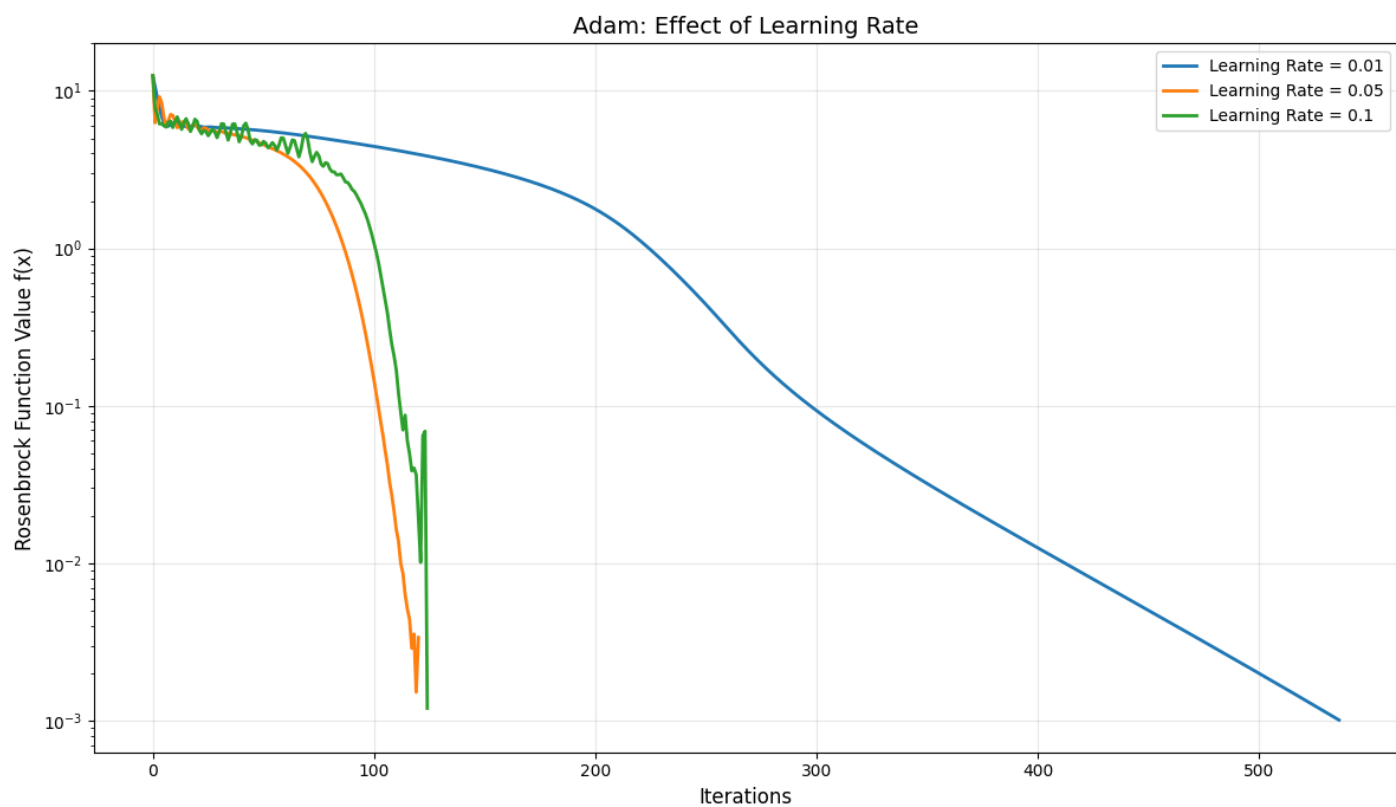
SGD with Momentum — Rosenbrock

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[0.900506966192946, 0.8104897543480012]	0.009917	627	0.023483	Converged
0.05	[0.9047283981270757, 0.8183671707458727]	0.009079	141	0.003723	Converged
0.10	[0.9280015286395668, 0.8545291560787335]	0.009616	135	0.003978	Converged



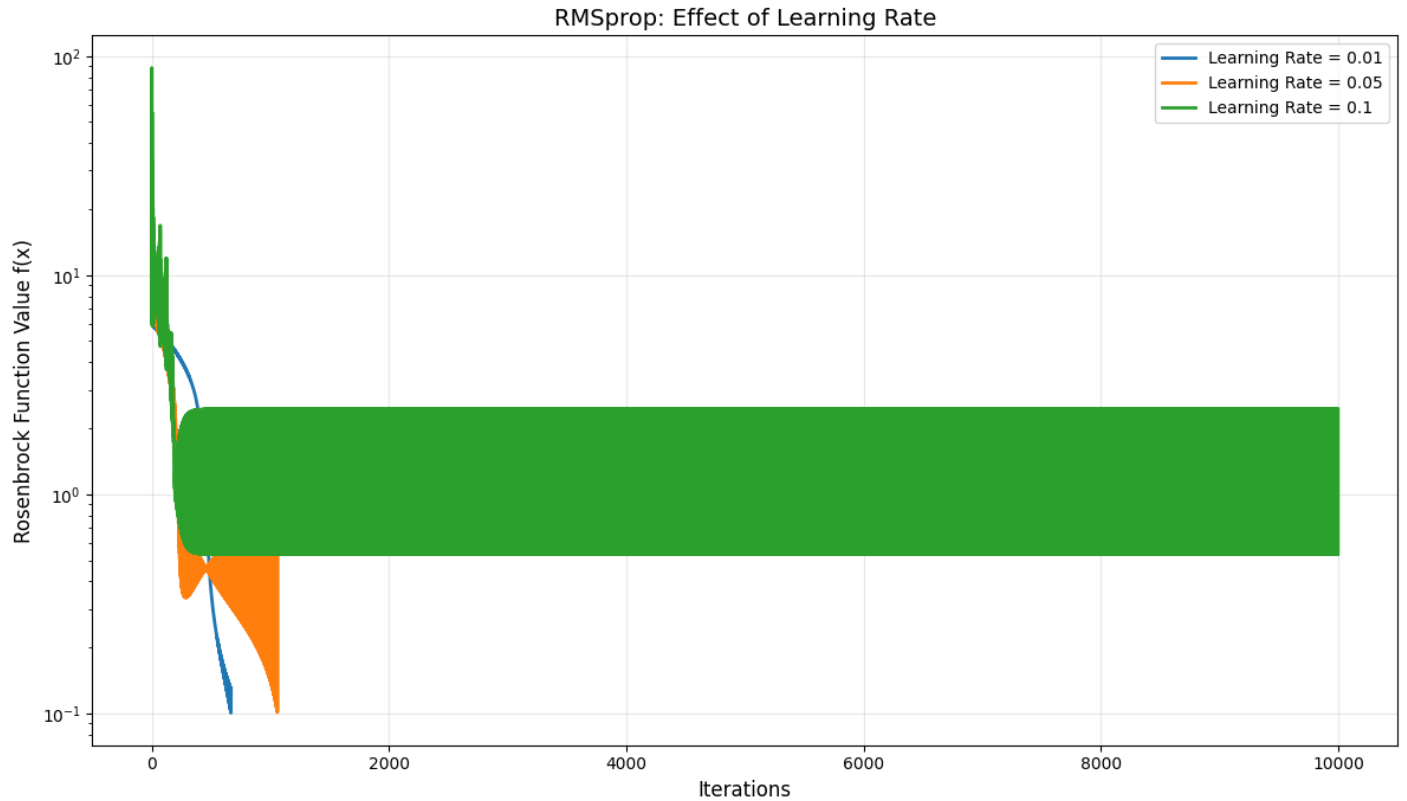
Adam — Rosenbrock

Learning Rate	Optimal x	Minimum $f(x)$	Iterations	Time (s)	Status
0.01	[0.9685201016596238, 0.9379006719100831]	0.000993	537	0.045241	Converged
0.05	[0.9831929886946519, 0.9688358754783347]	0.000752	121	0.007906	Converged
0.10	[0.9834958115645509, 0.9669236487985036]	0.000284	125	0.009164	Converged



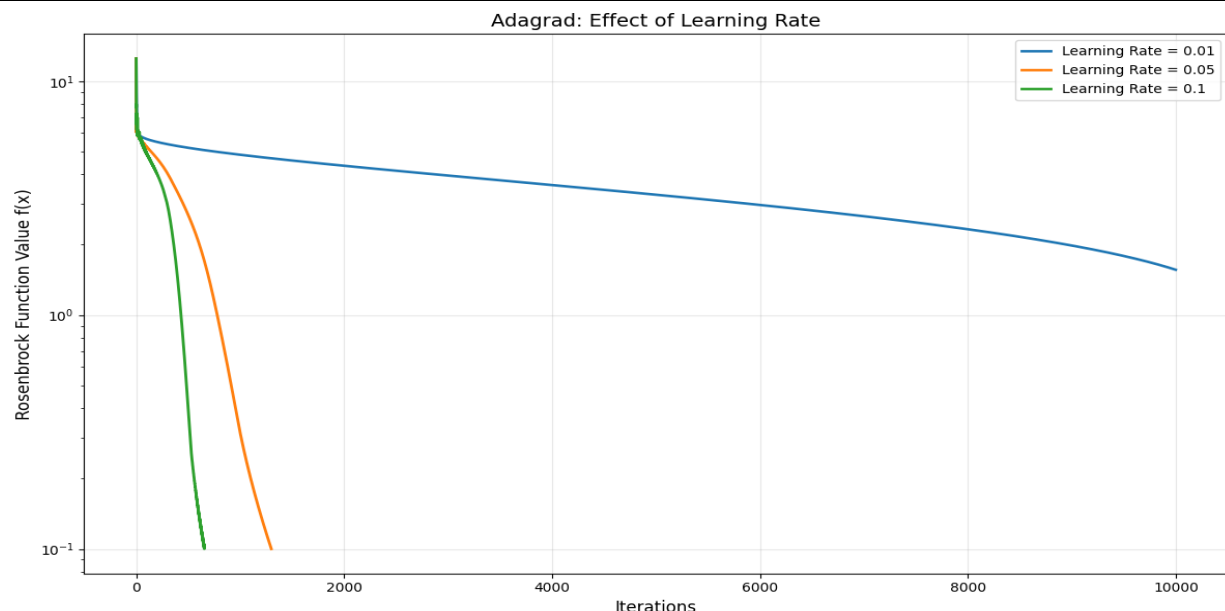
RMSprop — Rosenbrock

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[0.6926292761443033, 0.4728699313680588]	0.099190	670	0.029793	Converged
0.05	[0.7484815634402366, 0.5411987337398629]	0.099460	1061	0.032082	Converged
0.10	[0.3764308695249543, 0.1796133120542297]	0.532579	10000	0.310641	Maximum iterations reached



Adagrad — Rosenbrock

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[-0.23828613598197693, 0.07366627145365559]	1.561866	10000	0.343284	Maximum iterations reached
0.05	[0.6844891972272231, 0.46714450816462877]	0.099738	1301	0.036204	Converged
0.10	[0.6882313188255049, 0.4696541065991023]	0.098806	657	0.017774	Converged



6. Ackley Results

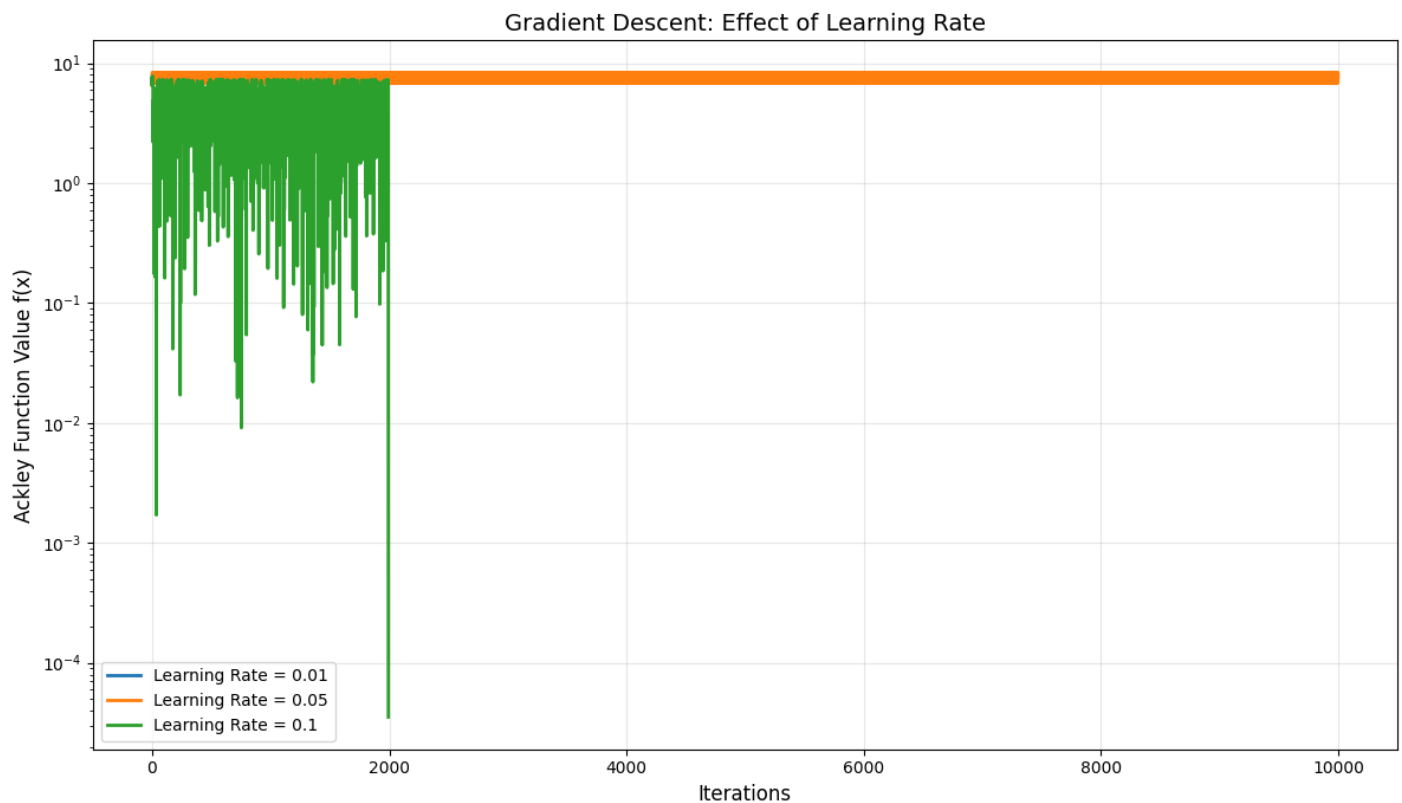
The notebook first evaluates the Ackley function from three initial guesses (-2.0, 0.0, and 1.0) using Nelder-Mead. The recorded results show that the local landscape depends strongly on the starting point: -2.0 leads to approximately $x=-1.9744$ and $f=6.5596$, 0.0 leads to $x=0$ and $f=0$, and 1.0 leads to approximately $x=0.9685$ and $f=3.5745$.

Gradient Descent — Ackley

Initial point = -2.0

Tolerance= 10^{-4}

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[-1.974451806067235]	6.559645375629525	4	0.000935	Converged
0.05	[-1.8857036748622735]	6.8784717049033155	10000	1.078003	Maximum iterations reached
0.10	[8.835854848954217e-06]	3.534757727097215e-05	1993	0.116552	Converged

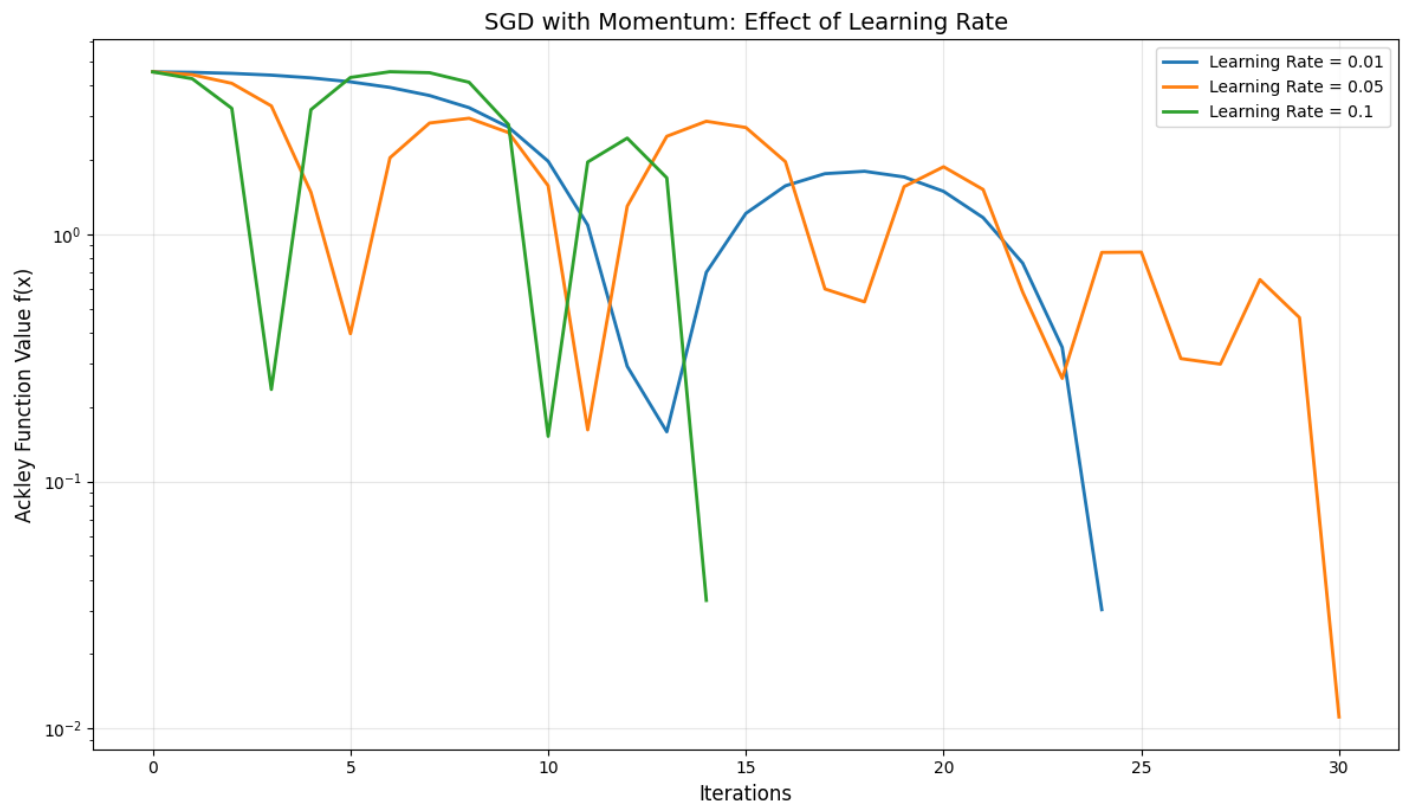


SGD with Momentum — Ackley

Initial point = -0.6

Tolerance= 10^{-1}

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[0.006930725807185421]	0.030279468720707303	25	0.008775	Converged
0.05	[-0.0026899873402019964]	0.011145280326758655	31	0.008128	Converged
0.10	[-0.007500600853509565]	0.03299635232177822	15	0.004013	Converged

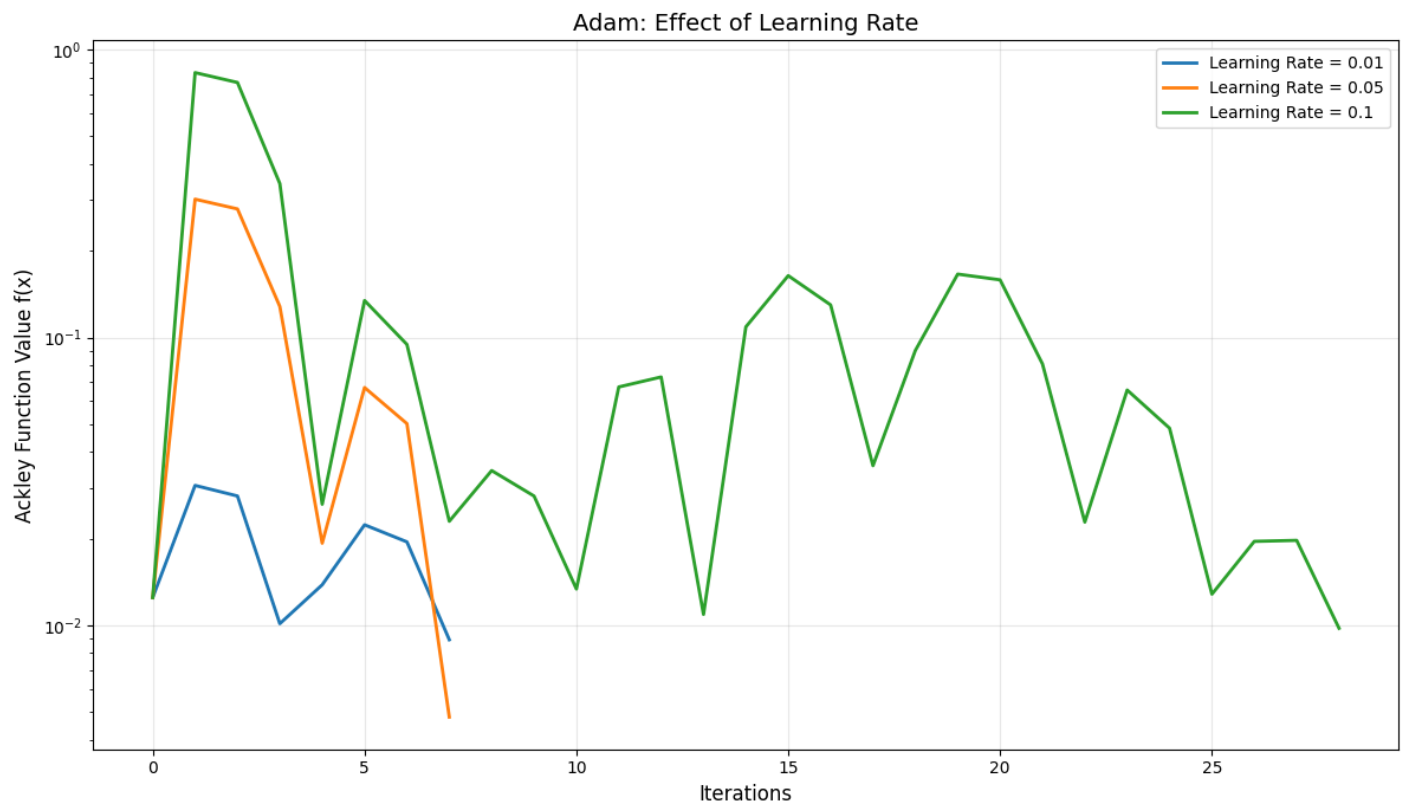


Adam — Ackley

Initial point = 0.003

Tolerance = 10^{-2}

Learning Rate	Optimal x	Minimum $f(x)$	Iterations	Time (s)	Status
0.01	[0.0021638443208169835]	0.00890472193234304	8	0.001641	Converged
0.05	[-.0011807783959150789]	0.004797364795117165	8	0.001539	Converged
0.10	[0.0023657466861903248]	0.00976102999294115	29	0.005526	Converged



RMSprop — Ackley

Initial point = 0.5

Tolerance= 10^{-1}

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[0.013912340346418554]	0.0659310447262027	36	0.005070	Converged
0.05	[-0.02499999996223095]	0.1330118043544748	10000	0.878243	Maximum iterations reached
0.10	[-0.04999999994435757]	0.32884220564281863	10000	0.655147	Maximum iterations reached

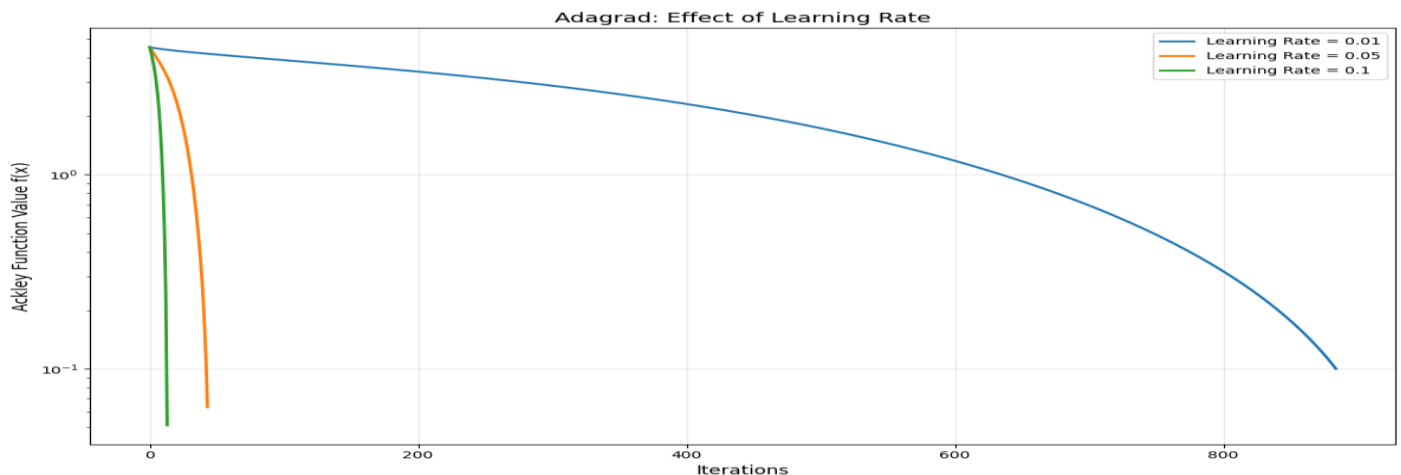


Adagrad — Ackley

Initial point = -0.6

Tolerance= 10^{-1}

Learning Rate	Optimal x	Minimum f(x)	Iterations	Time (s)	Status
0.01	[-.019792643586038897]	0.09992610805699487	885	0.172946	Converged
0.05	[-0.0134687991738923]	0.06351329453123755	44	0.007316	Converged
0.10	[-0.011146555522347917]	0.05119229284494908	14	0.005148	Converged



7. Discussion

The Rosenbrock experiments show clear differences among optimizers. SGD with Momentum and Adam require substantially fewer iterations than plain Gradient Descent for the 0.05 and 0.10 learning rates. Adam also achieves lower recorded objective values than RMSprop at the tested rates. RMSprop at 0.10 reaches the maximum iteration limit, while Adagrad at 0.01 also reaches the maximum iteration limit.

The Ackley experiments demonstrate stronger sensitivity to the starting point and local oscillatory structure. In the note recorded runs, SGD with Momentum and Adam converge quickly from their selected starting points. RMSprop converges at 0.01 but reaches the maximum iteration limit at 0.05 and 0.10. Adagrad converges for all three tested rates in its recorded run, with the 0.10 rate requiring the fewest iterations among the three.

A methodological limitation should be noted: the submitted notebook does not use one single Ackley starting point and one single convergence tolerance across every optimizer. For example, the recorded Ackley runs use different initial points in different cells, and the tolerance values differ between some optimizer functions. Therefore, the Ackley results should be interpreted as the results of the notebook as executed, rather than as a perfectly controlled head-to-head comparison.

8. Key Findings

- Rosenbrock's global optimum is correctly identified near (1,1).
- The simplified Ackley function has a global minimum at $x = 0$, but its oscillatory structure creates local minima.
- Learning rate has a substantial effect on both convergence speed and whether the maximum iteration limit is reached.
- Adam and SGD with Momentum generally show faster convergence than plain Gradient Descent in the recorded Rosenbrock experiments.
- RMSprop and Adagrad show more varied sensitivity to learning rate.
- For Ackley, the starting point is particularly important; the notebook's Part 1 Nelder-Mead experiment demonstrates this directly.

9. Conclusion

The notebook demonstrates the practical differences between several first-order optimization algorithms on two non-convex benchmark functions. Rosenbrock highlights the difficulty of a narrow curved valley, while Ackley highlights oscillatory local structure and sensitivity to initialization. The recorded experiments confirm that there is no universally best learning rate or optimizer: performance depends on the objective landscape and the optimizer's update mechanism. The results tables and convergence plots provide the basis for comparing iteration efficiency, final objective value, and convergence status.